

# PocketSphinx: Buffer overflows in language and acoustic model loading code

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|                       |                          |                                 |                      |
|-----------------------|--------------------------|---------------------------------|----------------------|
| 1<br>Total Advisories | 0<br>Critical            | 0<br>High                       | 1<br>Medium          |
| MEDIUM<br>Severity    | 5.5<br>CVSS / Risk Score | TLP:CLEAR<br>TLP Classification | PRO<br>Customer Tier |

## Executive Summary

### Impact The trie language model code introduced in PocketSphinx 5prealpha failed to check various boundary conditions when reading the headers of ARPA, DMP, and binary format language model files. In the case of invalid, corrupted or malicious input files, this could lead to stack and heap buffer overflows. In addition, the acoustic model loading code (which is over 30 years old...) contains numerous instances of `sscanf` with an unbounded string field which could also lead to stack overflows in the case of corrupt or malicious inputs. Because PocketSphinx will search the directory given by the `POCKETSPHINX\_PATH` environment variable for acoustic and language model files, if this directory is writable by untrusted users, an attacker could corrupt an existing file or write a malicious one to this directory in order to trigger the vulnerability. ### Patches The problem has been corrected in PocketSphinx 5.1.1. There is no patch currently available for users of Pocketsphinx 5pr

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## Top Threat Advisories

High-priority advisories ranked by CVSS score, exploitability, and actor attribution.

| Advisory Title                                                        | Severity | CVSS | EPSS | AI Summary                                                                          |
|-----------------------------------------------------------------------|----------|------|------|-------------------------------------------------------------------------------------|
| PocketSphinx: Buffer overflows in language and acoustic model loading | CRITICAL | 5.5  | -    | ### Impact<br><br>The trie language model code introduced in PocketSphinx 5prealpha |

## MITRE ATT&CK® v15 Mapping

Technique density score: 0.0 · Techniques observed: 0 · Tactics covered: 0

MITRE ATT&CK techniques pending analyst enrichment. Deploy detection rules from Section 6 to identify related activity.

## Threat Actor Intelligence

Attribution analysis across advisories in this report period.

| Threat Actor   | Count | Associated CVEs / Indicators |
|----------------|-------|------------------------------|
| CDB-UNATTR-CVE | 1     | -                            |

## Indicators of Compromise (IOCs)

IOC extraction for this advisory is pending enrichment pipeline completion. PRO subscribers receive real-time IOC push to SIEM/SOAR via the SENTINEL APEX webhook at: **GET /api/v1/intell/iocs?advisory={id}**. Full IOC packages include IPv4, domain, SHA256, and URL indicators with confidence scores, first-seen timestamps, and STIX 2.1 formatted bundles.

Detection Engineering

Deploy these production-ready detection rules into your SIEM platform to identify exploitation attempts in real time. Compatible with Microsoft Sentinel, Splunk ES, and any Sigma-compatible platform.

Sigma Rule - Webserver / Process Monitoring

```
title: SENTINEL APEX - Heap Overflow Exploitation Attempt
id: cdb-cdb-advisory-det
status: experimental
description: |
Detects exploitation attempts targeting CDB-ADVISORY (heap-based overflow).
Monitors for anomalous process crashes and memory allocation failures.
references:
- https://intel.cyberdudebivash.com
author: CyberDudeBivash SENTINEL APEX
date: 2026/07/18
tags:
- attack.initial_access
- attack.t1190
- attack.execution
- attack.t1203
logsource:
category: process_creation
product: windows
detection:
selection_crash:
EventID: 1000
ApplicationName|contains|any:
- 'PocketSphinx:'
selection_wer:
EventID: 1001
AppPath|contains|any:
- 'PocketSphinx:'
condition: selection_crash OR selection_wer
falsepositives:
- Legitimate software bugs / crash reporting
level: medium
```

Microsoft Sentinel - KQL Query

```
// SENTINEL APEX - PocketSphinx: Buffer overflows in language and acoustic model loading code
// Advisory | Microsoft Sentinel KQL
let time_window = ago(24h);
SecurityAlert
| where TimeGenerated > time_window
| where Description has "PocketSphinx: Buffer overflows in langua"
or AlertName has "PocketSphinx: Buffer overflows in langua"
or ExtendedProperties has "PocketSphinx: Buffer overflows in langua"
| extend Rule = "APEX-ADVISORY", Severity = "MEDIUM"
| project TimeGenerated, AlertName, AlertSeverity, Entities, Description, Rule, Severity
| order by TimeGenerated desc
```

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Incident Response Playbook

Phase 1 - 0-24 Hours (Containment)

- 1. Activate IR team; assign lead analyst and executive sponsor.
- 2. Isolate or WAF-shield PocketSphinx: Buffer overflows in language and acoustic model loading code instances exposed to the internet.
- 3. Enable verbose access logging on all PocketSphinx: Buffer overflows in language and acoustic model loading code endpoints immediately.
- 4. Pull last 72h of web server and application logs for forensic baselining.
- 5. Check SENTINEL APEX IOC feeds and SIEM alerts for exploitation hits in your environment.
- 6. Notify relevant internal stakeholders and legal / compliance team.

Phase 2 - 24-72 Hours (Eradication)

- 1. Apply vendor patch for PocketSphinx: Buffer overflows in language and acoustic model loading code or implement recommended mitigation.
- 2. Deploy Sigma and KQL detection rules from Section 6 across all SIEM platforms.
- 3. Rotate all API keys, service account credentials, and session tokens.
- 4. Conduct forensic timeline reconstruction for potential compromise window.
- 5. Validate patch deployment across 100% of affected instances via asset inventory.

Phase 3 - 7 Days (Recovery & Hardening)

- 1. Conduct post-incident review and update incident response runbooks.
- 2. Threat hunt for lateral movement or persistence artifacts using MITRE ATT&CK mapping.
- 3. Update asset inventory with patched version information and closure evidence.
- 4. Submit findings to internal risk register and CISO dashboard.
- 5. Schedule follow-up vulnerability scan to confirm remediation completeness.

Financial Impact Analysis (FAIR Model)

|                                                                          |                                                                       |                                                                       |                                                                |
|--------------------------------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------|----------------------------------------------------------------|
| <div><div>\$150K - \$900K</div><div>Breach Cost Range (FAIR)</div></div> | <div><div>\$380K</div><div>IBM Cost of Breach 2025 Median</div></div> | <div><div>\$120K/day</div><div>Business Interruption Cost</div></div> | <div><div>\$85K</div><div>Regulatory Fine Exposure</div></div> |
|--------------------------------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------|----------------------------------------------------------------|

Medium severity events require internal investigation and targeted customer notification. Average cost includes 340 hours of engineering time.

**Remediation Cost Estimate:** \$55K · Includes engineering time, IR retainer activation, forensic investigation, and customer notification costs.

Regulatory & Compliance Implications

Organizations processing this advisory must evaluate obligations under applicable regulatory frameworks. The table below maps this threat to relevant compliance requirements.

| Framework      | Reference     | Obligation                                                               |
|----------------|---------------|--------------------------------------------------------------------------|
| PCI-DSS v4.0   | Req. 6.3.3    | Non-critical patches applied within 3 months                             |
| NIST CSF 2.0   | ID.RA / PR.PS | Vulnerability assessment and platform security policies                  |
| ISO 27001:2022 | A.8.8         | Vulnerability management programme with documented remediation timelines |

Actionable Recommendations

Prioritized defensive actions based on this period's intelligence.

1. Monitor PocketSphinx: Buffer overflows in language and acoustic model loading code for escalation.
2. Apply patches in next maintenance window.
3. Apply vendor patch for PocketSphinx: Buffer overflows in language and acoustic model loading code immediately - check NVD for official fix guidance.
4. Enable enhanced logging and alerting on all systems running affected software versions.
5. Deploy the Sigma detection rule from this report into your SIEM platform (Sentinel / Splunk / Elastic).
6. Audit all internet-facing instances of affected products and confirm patch deployment.
7. Conduct a threat hunt using MITRE ATT&CK techniques mapped in this advisory.

Appendix - Report Metadata & Legal

| Field          | Value                                         |
|----------------|-----------------------------------------------|
| Report ID      | intel--580aab932f42f95915926798               |
| Report Type    | Weekly                                        |
| Platform       | CYBERDUDEBIVASH SENTINEL APEX v166.0 GOD-MODE |
| Generated At   | 2026-07-18T01:46:13.99916                     |
| Customer Tier  | PRO                                           |
| Customer Email | -                                             |
| TLP            | TLP: CLEAR                                    |
| STIX Version   | 2.1                                           |
| ATT&CK Version | v15                                           |

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